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# Step 1 -----
# Find runs of consecutive CpGs where the same parental hap is the
# meth outlier, was transmitted to the kid, and the kid hap on that
# side has ~the same meth level.
TOL = 0.10
df_meth_hits = (df_meth
  # sites where exactly one parental hap is the meth outlier
  .with_columns(
    n_hi = ((pl.col("dad_A") > 0.5)
      + (pl.col("dad_B") > 0.5)
      + (pl.col("mom_C") > 0.5)
      + (pl.col("mom_D") > 0.5)),
    outlier_hap = pl.when(pl.col("dad_A") > 0.5).then(pl.lit("A"))
      .when(pl.col("dad_B") > 0.5).then(pl.lit("B"))
      .when(pl.col("mom_C") > 0.5).then(pl.lit("C"))
      .otherwise(pl.lit("D")),
    outlier_meth = pl.max_horizontal("dad_A", "dad_B", "mom_C", "mom_D"),
  )
  .filter(pl.col("n_hi") == 1)
  # group consecutive CpGs sharing the same outlier hap
  .with_columns(
    run_id = (pl.col("outlier_hap")
      != pl.col("outlier_hap").shift(1)).cum_sum()
  )
  # collapse each run; outlier_meth = mean over its CpGs
  .group_by("run_id", maintain_order=True)
  .agg(
    chrom = pl.col("chrom").first(),
    start = pl.col("start").min(),
    end = pl.col("start").max(),
    n = pl.len(),
    outlier_hap = pl.col("outlier_hap").first(),
    outlier_meth = pl.col("outlier_meth").mean(),
    pat_hap = pl.col("pat_hap").first(),
    mat_hap = pl.col("mat_hap").first(),
    kid_pat = pl.col("kid_pat").mean(),
    kid_mat = pl.col("kid_mat").mean(),
  )
  .filter(pl.col("n") >= 3)
  # kid inherits the outlier hap with ~unchanged meth
  .filter(
    ((pl.col("outlier_hap") == pl.col("pat_hap"))
      & ((pl.col("kid_pat") - pl.col("outlier_meth")).abs() < TOL))
    | ((pl.col("outlier_hap") == pl.col("mat_hap"))
      & ((pl.col("kid_mat") - pl.col("outlier_meth")).abs() < TOL))
  )
)

# --- Step 2 -----
# Exactly one parental hap carries the ALT; kid inherits it on the
# matching side.
df_geno_hits = (df_geno
  # tag each SNV with the ALT count and the (unique) outlier hap
  .with_columns(
    n_alt = pl.sum_horizontal("dad_A", "dad_B", "mom_C", "mom_D"),
    outlier_hap = pl.when(pl.col("dad_A") == 1).then(pl.lit("A"))
      .when(pl.col("dad_B") == 1).then(pl.lit("B"))
      .when(pl.col("mom_C") == 1).then(pl.lit("C"))
      .otherwise(pl.lit("D")),
  )
  # keep SNVs with a single ALT-carrying parental hap
  .filter(pl.col("n_alt") == 1)
  # kid inherits that ALT on the matching side
  .filter(
    (pl.col("outlier_hap").is_in(["A", "B"]) & (pl.col("kid_pat") == 1))
    | (pl.col("outlier_hap").is_in(["C", "D"]) & (pl.col("kid_mat") == 1))
  )
)

# --- Step 3 -----
# Compound het = meth-outlier and geno-outlier from different
# parents, same locus.
df_compound_het = (df_meth_hits
  .rename({"outlier_hap": "outlier_hap_meth"})
  # pair each meth-outlier region with the nearest geno-outlier
  # SNV on the same chromosome, within 500 bp
  .join_asof(
    df_geno_hits.rename({"outlier_hap": "outlier_hap_geno"}),
    on="start", by="chrom",
    strategy="nearest", tolerance=500,
  )
  # keep loci where the meth and geno outliers are in trans
  # (i.e., come from different parents)
  .filter(
    (pl.col("outlier_hap_meth").is_in(["A", "B"])
      & pl.col("outlier_hap_geno").is_in(["C", "D"]))
  )
)

```